

RQ Answers

omitted

2024-04-05

Data overview

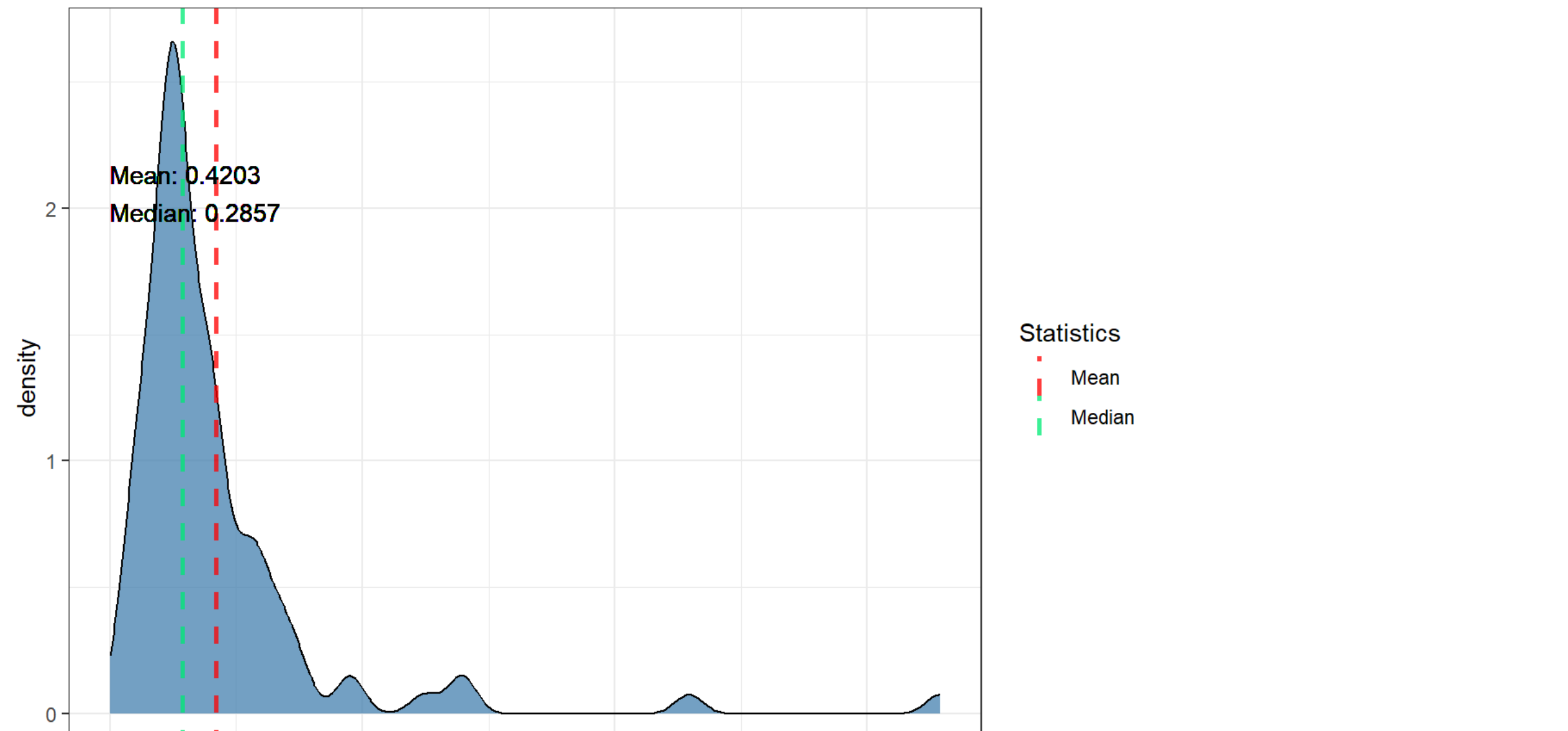
glimpse(all_data)				
## Rows: 3,509				
## Columns: 11				
## \$ repo	<chr> "commons-lang", "commons-lang", "commons-lang", "c...			
## \$ pr_number	<int> 1001, 1001, 1001, 1001, 1001, 1001, 1001, 10...			
## \$ rule	<chr> "java:S127", "java:S1133", "java:S3776", "java:S37...			
## \$ severity	<chr> "MAJOR", "INFO", "CRITICAL", "CRITICAL", "INFO", "...			
## \$ file	<chr> "src/main/java/org/apache/commons/lang3/StringUtil...			
## \$ type	<chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...			
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...			
## \$ debt	<int> 10, 10, 11, 11, 10, 10, 10, 10, 2, 2, 16, 10, ...			
## \$ ncloc_affected_file	<int> 2780, 2780, 2780, 2780, 2780, 2780, 2780, 27...			
## \$ complexity	<int> 964, 964, 964, 964, 964, 964, 964, 964, 964, ...			
## \$ origin	<chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 93 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 93 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

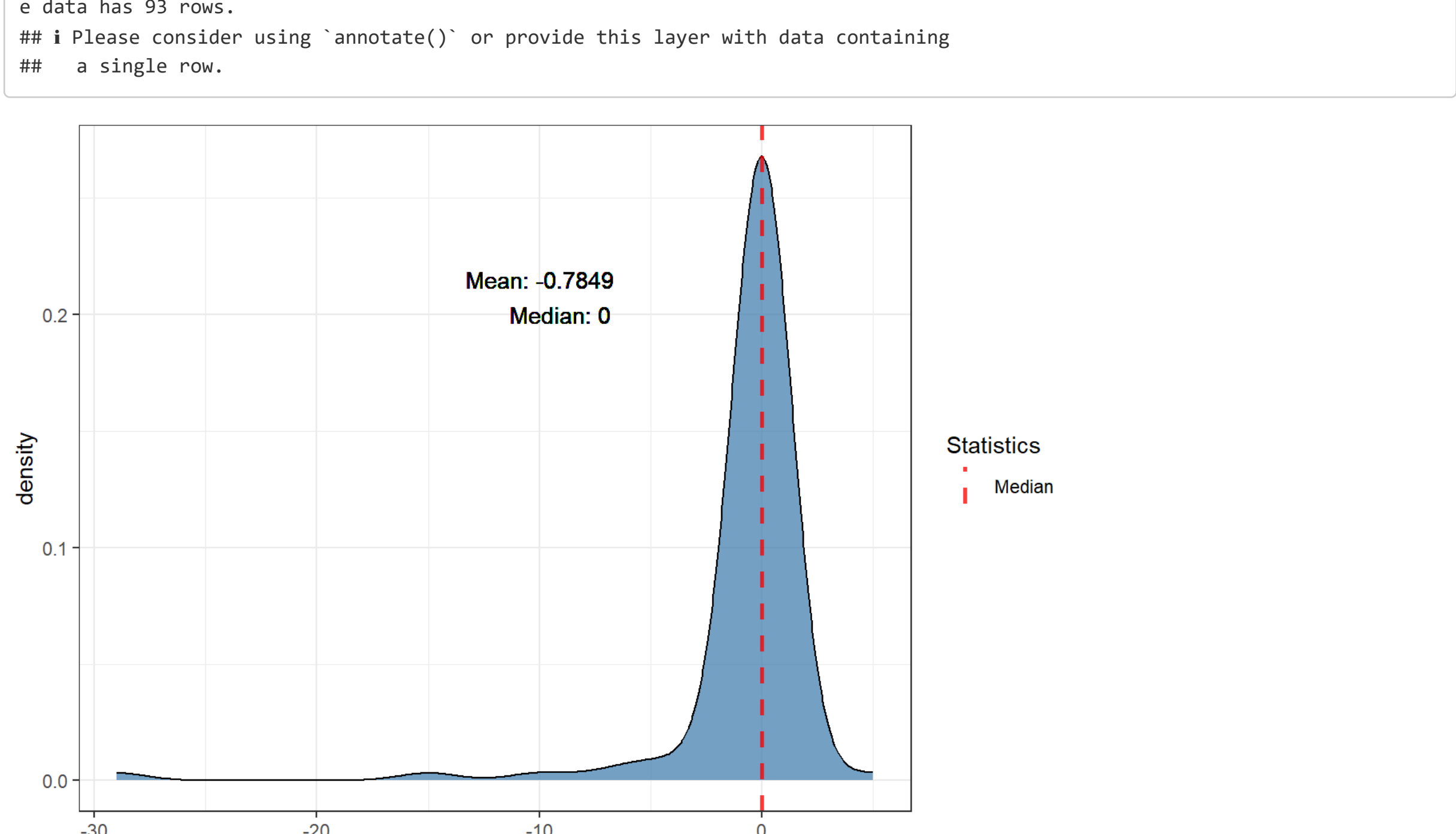


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 93 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 93 rows.
## | Please consider using `annotate()` or provide this layer with data containing
## | a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	<dbl>
## 1 commons-lang	9.68	89.2	1.08	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	<dbl>
## 1	9.68	89.2	1.08

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>
## 1 commons-lang	90.3	9.68	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	90.3	9.68

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>
## 1 commons-lang	100	0	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>
## 1	100	0

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a table of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank `commons-lang`		
## <int> <chr>		
## 1	1 java:S3358	- 5
## 2	2 java:S2293	- 3
## 3	3 java:S1155	- 3
## 4	4 java:S2326	- 2
## 5	5 java:S1066	- 2
## 6	6 java:S1854	- 1
## 7	7 java:S1125	- 1
## 8	8 java:S1126	- 1
## 9	9 java:S117	- 1
## 10	10 java:S1135	- 1

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S3358	122	
## 2 java:S1155	123	
## 3 java:S2293	124	
## 4 java:S1066	125	
## 5 java:S2326	126	
## 6 java:S1125	127	
## 7 java:S1126	128	
## 8 java:S1135	129	
## 9 java:S117	130	
## 10 java:S1854	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank `commons-lang`		
## <int> <chr>		
## 1	1 java:S1168	- 694
## 2	2 java:S1133	- 526
## 3	3 java:S3776	- 381
## 4	4 java:S1149	- 250
## 5	5 java:S1905	- 144
## 6	6 java:S1192	- 135
## 7	7 java:S127	- 113
## 8	8 java:S135	- 107
## 9	9 java:S1118	- 105
## 10	10 java:S2259	- 90

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1168	122	
## 2 java:S1133	123	
## 3 java:S3776	124	
## 4 java:S1149	125	
## 5 java:S1905	126	
## 6 java:S1192	127	
## 7 java:S127	128	
## 8 java:S135	129	
## 9 java:S1118	130	
## 10 java:S2259	131	

Outliers

Higher TDD

## # A tibble: 93 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 commons-lang	333	470	143	3.29
## 2 commons-lang	351	360	157	2.29
## 3 commons-lang	813	400	285	1.40
## 4 commons-lang	610	1434	1035	1.39
## 5 commons-lang	1129	757	609	1.24
## 6 commons-lang	924	90	94	0.957
## 7 commons-lang	870	400	424	0.943
## 8 commons-lang	670	284	374	0.759
## 9 commons-lang	925	264	354	0.746
## 10 commons-lang	1085	147	215	0.684
## # i 83 more rows				

Higher TDV

## # A tibble: 93 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	407 commons-lang	5	
## 2	214 commons-lang	0	
## 3	216 commons-lang	0	
## 4	237 commons-lang	0	
## 5	266 commons-lang	0	
## 6	302 commons-lang	0	
## 7	312 commons-lang	0	
## 8	333 commons-lang	0	
## 9	351 commons-lang	0	
## 10	380 commons-lang	0	
## # i 83 more rows			

Lower TDV

## # A tibble: 93 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	334 commons-lang	-29	
## 2	800 commons-lang	-15	
## 3	403 commons-lang	-10	
## 4	1033 commons-lang	-7	
## 5	680 commons-lang	-5	
## 6	941 commons-lang	-5	
## 7	348 commons-lang	-3	
## 8	701 commons-lang	-2	
## 9	1059 commons-lang	-2	
## 10	214 commons-lang	0	
## # i 83 more rows			

Extra

Number of PRs with pre-existing TD

n
1 93

Number of java:S1192 issues that affect test code

n
1 0

Number of java:S1192 issues that affect not test code

n
1 135

Summary NCLOC

ncloc
Min. : 34
1st Qu.: 304
Median : 615
Mean : 1608
3rd Qu.: 2806
Max. : 16707